

API Integration Assignment

Selected API

I selected the Open-Meteo Weather API, a free public API that provides current weather and forecast data without requiring authentication.

API Endpoint

https://api.open-meteo.com/v1/forecast?latitude=23.99&longitude=90.23¤t=temperature_2m

This endpoint returns the current temperature for the specified latitude and longitude.

JavaScript Implementation

```
async function getWeather() {
  try {
    const response = await fetch(

"https://api.open-meteo.com/v1/forecast?latitude=23.99&longitude=90.23&current=temperature_2m"
    );

    const data = await response.json();

    console.log(
      "Current Temperature:",
      data.current.temperature_2m + " °C"
    );
  } catch (error) {
    console.error("Error:", error);
  }
}

getWeather();
```

Response Format

The API returns data in JSON format.

Example response:

```
{
  "current": {
    "temperature_2m": 29.7
  }
}
```

The value used in this project is:

`data.current.temperature_2m`

which contains the current temperature.

Displayed Data Point

The selected data point is the current temperature, which is displayed in the browser console.

Real-World Use Case

This API could be used to build a weather application that displays current weather conditions and forecasts for different locations. It could also be integrated into travel, agriculture, or outdoor activity planning applications to provide users with real-time weather information.

Learning Outcomes

Through this project, I learned how APIs provide dynamic data to web applications, how to retrieve data using JavaScript's `fetch()` method, how to interpret JSON responses, and how API integration enables responsive, data-driven web experiences.